Observable martingales on Brownian-time paths: predictable energy, associativity and uniform reconstruction

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Reviewed editorial draft, revision 3, October 29, 2026

Abstract

On a constructed Brownian-time Brownian probability space, predictable integrals against terminal-observable martingales admit an exact change of energy measure. This gives a literal product isometry on the entire output-energy L^2 space and associativity without boundedness of the second integrand. We then retain all signed cross-channel covariances, pass through their null quotient and completion, and identify every common bounded predictable action with the original output-channel integral. A fixed countable family of actual integrals reconstructs continuous curves of observables uniformly on compact parameter sets, simultaneously over all causal horizons. Original native graph tests identify the family's scalar coefficients, with a quantitative graph-dual norm bound. A bounded two-source Gaussian recovery transfers this convergence to the original intrinsic divergence graph. The proofs separate raw predictability, right-germ martingales and source-operator domains. These are classical integration and Hilbert-space mechanisms assembled as a formalization companion; they do not provide a temporal Itô or Clark–Ocone calculus for the uncorrected Brownian-time path.

1 Probability space and the two filtrations

Let B and W be independent standard Brownian motions constructed on two independent countable Gaussian source spaces. The process considered here is

$$Z_t = B_{|W_t|}, \qquad t \geq 0.$$

The outer Brownian motion is one-sided and the clock is reflected. Thus this paper concerns this particular Brownian-time Brownian motion; terminology for two-sided versions of iterated Brownian motion does not change the process used below. Write $(\Omega, \mathcal{A}, \mathbb{P})$ for the joint source space, $\mathcal{H} = L^2(\Omega, \mathcal{A}, \mathbb{P}; \mathbb{R})$, and

$$\mathcal{F}_t^Z = \sigma(Z_s : 0 \leq s \leq t), \qquad \mathcal{F}_\infty^Z = \sigma(Z_s : s \geq 0), \qquad \mathcal{O} = L^2(\mathcal{F}_\infty^Z) \subset \mathcal{H}.$$

All L^2 spaces consist of equivalence classes. Statements about measurability of such a class mean that it has a representative with the stated measurability.

The raw filtration $\mathcal{F}^Z$ is used for integrands. Let $\mathcal{P}$ be its predictable sigma algebra on $S = [0, \infty) \times \Omega$. It is generated by

$$(s, t] \times A, \quad 0 \leq s \leq t < \infty, \quad A \in \mathcal{F}_s^Z, \qquad \{0\} \times A, \quad A \in \mathcal{F}_0^Z. \tag{1}$$

For martingales use the right-continuous ambient-null augmentation

$$\mathcal{G}_t = \bigcap_{u > t} (\mathcal{F}_u^Z \vee \mathcal{N}), \qquad P_t f = \mathbb{E}[f \mid \mathcal{G}_t],$$

where $\mathcal{N}$ denotes ambient measurable null sets and their ambient measurable subsets. Passing to the completion of the probability space when applying a classical regularization theorem leaves the L^2 objects unchanged. The frozen construction realizes P_t as orthogonal projection onto

$$\bigcap_{u>t} L^2(\mathcal{F}_u^Z).$$

It proves the identification with $L^2(\mathcal{G}_t)$, rather than assuming that an intersection of raw sigma algebras has the desired representatives. In particular

$$P_s P_t = P_{\min(s,t)}, \quad \|P_t f\| \le \|f\|, \quad P_t f \in \mathcal{O}, \quad P_n f \longrightarrow f \quad (f \in \mathcal{O}). \tag{2}$$

The last assertion follows also from increasing closed-subspace projection convergence: every raw finite past is contained in its right germ, and their union generates $\mathcal{F}_\infty^Z$ modulo null sets.

For a fixed $F \in \mathcal{H}$, $M_t^F = P_t F$ is a square-integrable $\mathcal{G}$-martingale. It has right-continuous L^2 dependence on time. This fixed-terminal martingale differs from Z; no martingale assumption on Z enters any construction. Write

$$G_0 = P_0, \qquad \mathcal{O}_0 = \mathcal{O} \cap \ker G_0.$$

We retain the whole initial germ in the statements, even where particular observables have zero initial projection.

2 The original predictable integral

We first specify the energy measure and integral to which later associativity statements refer. They are established objects, not definitions imposed by the desired change-of-integrator identity.

Proposition 2.1 (Finite predictable energy)**.** *For every $F \in \mathcal{H}$ there is a unique finite measure μ_F on $(S, \mathcal{P})$ satisfying*

$$\mu_F((s,t] \times A) = \mathbb{E}\mathbf{1}_A (M_t^F - M_s^F)^2, \quad A \in \mathcal{F}_s^Z, \tag{3}$$

$$\mu_F(\{0\} \times A) = 0, \qquad \mu_F(S) \le \|F\|^2. \tag{4}$$

Finite linear combinations of rectangle indicators are dense in $L^2(\mathcal{P}, \mu_F)$, with origin indicators representing zero.

Proof. The existence assertion is the finite Doléans-measure construction for a square-integrable martingale, restricted to the raw predictable sigma algebra. For precision, on the usual completion of $\mathcal{G}$ take a càdlàg modification of M^F and the predictable compensator A^F of $(M^F)^2$, normalized by $A_0^F = 0$. The classical martingale regularization and compensator theorems apply: M^F is closed by F, and its stopped squares are uniformly integrable, since they are bounded by conditional expectations of the integrable variable F^2. On each finite horizon $(M^F)^2 - A^F$ is a martingale and

$$\mathbb{E}A_T^F = \|P_T F\|^2 - \|P_0 F\|^2 \le \|F\|^2.$$

The positive measure $\mathbb{E}\int_{(0,\infty)} \mathbf{1}_E(r,\omega)\, dA_r^F(\omega)$, restricted to $\mathcal{P}$, is therefore finite. For $A \in \mathcal{F}_s^Z \subset \mathcal{G}_s$, conditional martingale identities give

$$\mathbb{E}\mathbf{1}_A (A_t^F - A_s^F) = \mathbb{E}\mathbf{1}_A ((M_t^F)^2 - (M_s^F)^2) = \mathbb{E}\mathbf{1}_A (M_t^F - M_s^F)^2.$$

This verifies the required rectangle values. The compensator here belongs only to the auxiliary martingale M^F.

The frozen proof instead constructs the raw rectangle content directly, proves continuity at the empty set by compact-fiber approximation, and extends that content. The classical construction just given has the same values, so uniqueness identifies the measures. Indeed, rectangles and origin sets form a generating semiring, and a countable increasing family $(0, n] \times \Omega$, together with the origin, covers S. Finite-measure uniqueness applies. The span of its indicators is L^2 dense by the monotone class theorem and truncation; the origin has zero mass. This also proves the asserted density for the directly constructed formal measure. Classical measure and regularization background can be found in [1]. □

Proposition 2.2 (Isometric extension)**.** *There is a unique linear isometry*

$$I_F : L^2(\mathcal{P}, \mu_F) \longrightarrow \mathcal{H}$$

which on a raw predictable rectangle has value

$$I_F(\mathbf{1}_{(s,t]\times A}) = \mathbf{1}_A(P_t - P_s)F. \tag{5}$$

Its range is contained in $\mathcal{O}_0$. For $H \in L^2(\mu_F)$,

$$I_F(\mathbf{1}_{(0,r]}H) = P_r I_F(H), \qquad r \geq 0. \tag{6}$$

Time indicators in this paper include the entire sample space.

Proof. Refine a finite collection of rectangles by all of their time endpoints. The resulting process has the form $H = \sum_j h_j \mathbf{1}_{(t_{j-1},t_j]}$, where each h_j is a finite linear combination of $\mathcal{F}^Z_{t_{j-1}}$ event indicators. Assign the sum $\sum_j h_j(M^F_{t_j} - M^F_{t_{j-1}})$. Distinct time-cell outputs are orthogonal: the earlier one is $\mathcal{G}_{t_{j-1}}$ measurable and the later increment has zero conditional expectation there. Within one cell, partition the event indicators into disjoint atoms. Formula (3) then gives

$$\left\| \sum_j h_j(M^F_{t_j} - M^F_{t_{j-1}}) \right\|^2 = \int_S H^2 \, d\mu_F.$$

In particular null representatives have zero output. Refinement changes neither process nor output, so this defines an isometry on the rectangle span, and density gives its unique extension.

Each finite output is observable at some finite raw horizon strictly larger than all the endpoints: $P_t F$ is measurable modulo null sets at every raw time $u > t$. Its conditional expectation at $\mathcal{G}_0$ is zero by the martingale identity and raw adaptedness of h_j. Closedness of $\mathcal{O}$ and of $\ker G_0$ passes both properties to the limit. Finally, conditional expectation of a single cell output at $\mathcal{G}_r$ is zero for $r \leq s$, is $\mathbf{1}_A(P_r - P_s)F$ for $s < r < t$, and is the whole output for $r \geq t$. These are exactly the integrals of its time cutoff. Linearity and the contractivity of cutoff and P_r prove (6) for every H. □

Lemma 2.3 (Raw rectangle pull-out)**.** *If $Y = I_F(H)$ and $A \in \mathcal{F}^Z_s$, then*

$$I_F(\mathbf{1}_{(s,t]\times A}H) = \mathbf{1}_A(P_t - P_s)Y. \tag{7}$$

Proof. First consider the future mask $(s,\infty)\times A$ and one input cell $(a,b]\times C$. If $b\le s$, its product is zero, as is $\mathbf{1}_A(I-P_s)$ of the cell output. Otherwise replace its left endpoint by $\ell=\max(a,s)$, and replace its event by $A\cap C$. The event belongs to $\mathcal{F}^Z_\ell$, and the integral of the masked cell is

$$\mathbf{1}_{A\cap C}(P_b-P_\ell)F=\mathbf{1}_A(I-P_s)(\mathbf{1}_C(P_b-P_a)F).$$

The equality follows by splitting $s\le a$ and $a<s<b$ and applying conditional expectation. It extends from the cell span to all H: both mask multiplication in $L^2(\mu_F)$ and the output operators $\mathbf{1}_A(I-P_s)$ are bounded. Subtract the analogous future identity at t, using $A\in\mathcal{F}^Z_t$. The result is (7). □

3 Exact change of energy and literal products

The pull-out lemma compares the original integral with the original energy measure of its actual output. It supplies the domain information needed for an unbounded second integrand.

Theorem 3.1 (Energy transformation)**.** *For $F\in\mathcal{H}$, $H\in L^2(\mathcal{P},\mu_F)$ and $Y=I_F(H)$,*

$$\mu_Y=H^2\mu_F \tag{8}$$

as finite measures on the whole raw predictable sigma algebra. Here and below products are formed from predictable measurable representatives.

Proof. The right side is a finite measure of mass $\|H\|^2_{L^2(\mu_F)}$. For a raw rectangle $R=(s,t]\times A$, the isometry and Lemma 2.3 give

$$\int_R H^2\,d\mu_F=\|I_F(\mathbf{1}_RH)\|^2=\mathbb{E}\mathbf{1}_A((P_t-P_s)Y)^2=\mu_Y(R).$$

Both measures vanish on origin sets. The generating-semiring uniqueness argument from Proposition 2.1 proves equality on $\mathcal{P}$. □

Theorem 3.2 (Full-domain associativity)**.** *With F,H,Y as above, multiplication defines a linear isometry*

$$T_H:L^2(\mathcal{P},\mu_Y)\longrightarrow L^2(\mathcal{P},\mu_F),\qquad T_HK=HK.$$

For every K in this entire output-energy space,

$$\|T_HK\|_{L^2(\mu_F)}=\|K\|_{L^2(\mu_Y)},\qquad I_Y(K)=I_F(T_HK). \tag{9}$$

There is no boundedness requirement on H or K beyond their stated energy domains.

Proof. The density identity gives

$$\int H^2K^2\,d\mu_F=\int K^2\,d\mu_Y<\infty.$$

It also handles the null ideals. If $K=L$ outside a μ_Y-null set N, then $\int_N H^2\,d\mu_F=0$, so $H=0$ for μ_F almost every point of N. Hence $HK=HL$ in $L^2(\mu_F)$. Changing the representative of H changes neither the density measure nor the product class. Additivity and homogeneity hold for measurable representatives and descend to classes. The displayed equality proves the norm identity.

For $K=\mathbf{1}_{(s,t]\times A}$, the cell formula for I_Y and the pull-out lemma show

$$I_Y(K)=\mathbf{1}_A(P_t-P_s)Y=I_F(\mathbf{1}_{(s,t]\times A}H).$$

Origin indicators have zero class. Both I_Y and I_FT_H are continuous isometries on the same $L^2(\mu_Y)$ space, and they agree on its dense rectangle span. This proves (9). □

The product theorem concerns two different energy measures. For example, if H vanishes on a set of positive μ_F mass, μ_Y forgets that set, and multiplication by H removes the ambiguity there. No reverse absolute continuity is asserted. This is not an assertion that products of arbitrary $L^2(\mu_F)$ functions remain in $L^2(\mu_F)$. The associativity and domain-equivalence mechanism is classical; for a continuous-martingale comparison see [2, Proposition 5.4]. Continuity of the auxiliary IBM-observable martingales is not assumed here.

4 Signed joint energy and common multipliers

A family of channels must retain their cross covariances. Let $\mathbf{F} = (F_i)_{i\in\mathcal{I}}$ be any family in $\mathcal{H}$. Its raw input space is the algebraic direct sum

$$\mathcal{R}_{\mathbf{F}} = \bigoplus_{i\in\mathcal{I}} L^2(\mathcal{P}, \mu_{F_i}), \qquad J_{\mathrm{raw}} h = \sum_i I_{F_i}(h_i).$$

Only finitely many components of a raw input are nonzero. Define

$$\langle h, k\rangle_{\mathrm{en}} = \langle J_{\mathrm{raw}} h, J_{\mathrm{raw}} k\rangle_{\mathcal{H}}\,, \qquad \mathcal{N}_{\mathbf{F}} = \ker J_{\mathrm{raw}}. \tag{10}$$

Quotient by $\mathcal{N}_{\mathbf{F}}$ and complete in this inner product. Denote the resulting Hilbert space by $\mathcal{C}_{\mathbf{F}}$. The actual output extends uniquely to an isometry $J_{\mathbf{F}} : \mathcal{C}_{\mathbf{F}} \to \mathcal{H}$ with closed range in $\mathcal{O}_0$. This is not the direct sum norm obtained by discarding cross terms.

For completeness those cross terms are genuine signed measures. Set

$$\nu_{FG} = \tfrac{1}{4}(\mu_{F+G} - \mu_{F-G}).$$

The linearity of P_t and (3) give

$$\nu_{FG}((s,t]\times A) = \mathbb{E}\mathbf{1}_A((P_t - P_s)F)((P_t - P_s)G).$$

The same refinement argument as for the isometry proves the pairing formula on rectangle-step inputs. Its extension is justified by the following measure estimate.

Lemma 4.1 (Cross-energy integrability)**.** *For $H \in L^2(\mu_F)$ and $K \in L^2(\mu_G)$,*

$$\int |HK|\, d|\nu_{FG}| \le \|H\|_{L^2(\mu_F)}\, \|K\|_{L^2(\mu_G)}\,, \qquad \langle I_F H, I_G K\rangle = \int HK\, d\nu_{FG}. \tag{11}$$

Proof. Rectangle values and finite signed-measure uniqueness first give

$$\mu_{aF+bG} = a^2\mu_F + 2ab\nu_{FG} + b^2\mu_G \quad (a, b \in \mathbb{R}).$$

In particular $\mu_F + \mu_G \pm 2\nu_{FG}$ are positive. Consequently ν_{FG}, and hence its variation, is absolutely continuous with respect to $\lambda = \mu_F + \mu_G$: on a λ-null set the two positive-measure inequalities force its signed value on every subset to vanish. Write $a = d\mu_F/d\lambda$, $b = d\mu_G/d\lambda$ and $c = d\nu_{FG}/d\lambda$. Positivity of μ_{F+qG} says $a + 2qc + q^2 b \ge 0$ almost everywhere for each rational q. A common exceptional set suffices for all rationals, and continuity gives the inequality for all real q. Its discriminant yields $c^2 \le ab$ almost everywhere. The variation density is $v = |c|$. Thus

$$\int |HK| v\, d\lambda \le \int |H|\sqrt{a}\, |K|\sqrt{b}\, d\lambda \le \Big(\int H^2\, d\mu_F\Big)^{1/2}\Big(\int K^2\, d\mu_G\Big)^{1/2}.$$

Approximate H, K separately by rectangle-step functions in their own energy norms. The bound just proved controls both product-error terms in $L^1(|\nu_{FG}|)$. Their actual integral outputs converge by isometry. Passing to the limit in the finite pairing identity proves the second formula. □

Theorem 4.2 (Bounded common action)**.** *Let $g : S \to \mathbb{R}$ be raw predictable and pointwise bounded by $C \geq 0$. Multiplication of every component by this same g descends to a bounded linear operator M_g on $\mathcal{C}_{\mathbf{F}}$ with $\|M_g\| \leq C$. For all z in that completion, setting $Y = J_{\mathbf{F}} z$ gives*

$$J_{\mathbf{F}}(M_g z) = I_Y([g]_{\mu_Y}). \tag{12}$$

Proof. Put $g^{\perp} = (C^2 - g^2)^{1/2}$, a bounded predictable function. For raw h, Lemma 4.1 and $g^2 + (g^{\perp})^2 = C^2$ imply the exact partition identity

$$\|J_{\mathrm{raw}}(gh)\|^2 + \left\|J_{\mathrm{raw}}(g^{\perp}h)\right\|^2 = C^2 \|J_{\mathrm{raw}}h\|^2 .$$

Both squared norms on the left are nonnegative, including every signed cross term. Hence $\|J_{\mathrm{raw}}(gh)\| \leq C \|J_{\mathrm{raw}}h\|$. This proves preservation of the null relation and extension to the completion.

We next give the single-channel representation needed to identify the extension. For any $f \in \mathcal{O}$, the rectangle formula yields

$$I_f(\mathbf{1}_{(0,n]}) = P_n f - P_0 f \longrightarrow f - P_0 f.$$

The isometric range of I_f is closed, so some $H_f \in L^2(\mu_f)$ has output $f - P_0 f$. Since $Y \in \mathcal{O}_0$, this gives $K \in L^2(\mu_Y)$ with $I_Y K = Y$. This is a representation by a channel indexed by the target itself; it asserts no predictable representation property for a single fixed driving martingale.

Append the channel Y to the family. The same common-multiplier bound applied to a raw family input h minus the appended input K gives

$$\|J_{\mathrm{raw}}(gh) - I_Y(gK)\| \leq C \|J_{\mathrm{raw}}h - Y\| .$$

Approximate z by raw inputs and pass to the limit. It follows that $J_{\mathbf{F}}(M_g z) = I_Y(gK)$. Theorem 3.2, applied to $I_Y K = Y$, now gives $I_Y(gK) = I_Y([g]_{\mu_Y})$. Here the right-hand integral uses the original energy measure of the output Y; $[g]$ is admissible because that measure is finite. This proves (12). □

The hypothesis that g is common to all channels is essential. If two channels are identical, the raw pair $(H, -H)$ has zero joint energy. Multiplying only its first component by one and its second by zero usually gives a nonzero output. Thus unrelated channelwise multipliers need not even descend to the quotient. This elementary example explains why diagonal energy bounds alone would not prove the theorem.

5 A fixed countable stochastic family

The preceding target-indexed representation yields a fixed countable family on the separable observable space. The joint Gaussian product has separable L^2, and $\mathcal{O}_0$ is a closed subspace. Choose a countable Hilbert basis $(e_i)_{i \in I}$ of $\mathcal{O}_0$, permitting a finite or empty index set when appropriate. Regard each e_i as a terminal channel and choose $h_i \in L^2(\mu_{e_i})$ with

$$I_{e_i}(h_i) = e_i, \qquad \|h_i\| = 1. \tag{13}$$

Existence follows from the self-representation argument because $P_0 e_i = 0$. These are actual raw predictable integrands. Their construction uses a Hilbert basis and closed ranges; no explicit causal gradient formula is asserted.

For the family $\mathbf{e} = (e_i)$, $J = J_{\mathbf{e}}$ maps the completed space $\mathcal{C}_{\mathbf{e}}$ isometrically onto $\mathcal{O}_0$: its closed range contains every e_i. For $f \in \mathcal{O}$ set

$$r(f) = f - P_0 f, \quad c_i(f) = \langle e_i, f \rangle, \quad L(f) = J^{-1} r(f), \quad Q_A(f) = \sum_{i \in A} [\iota_i(c_i(f) h_i)],$$

where $A \subset I$ is finite and ι_i inserts a single raw channel. The bracket means its image in the null quotient and completion. Parseval gives both

$$JQ_A(f) = \sum_{i \in A} c_i(f) e_i, \qquad \|Q_A(f) - L(f)\|^2 = \|r(f)\|^2 - \sum_{i \in A} |c_i(f)|^2. \tag{14}$$

The maps r, L and c_i are continuous and linear.

Stopping is the contraction

$$S_r = J^{-1} P_r J \quad \text{on } \mathcal{C}_{\mathbf{e}}.$$

It is well-defined since P_r preserves $\mathcal{O}_0$. On a single input it equals literal time cutoff, by (6) and injectivity of J. Define the finite actual output

$$U_{A,r}(f) = \sum_{i \in A} I_{e_i}(\mathbf{1}_{(0,r]} c_i(f) h_i) = JS_r Q_A(f). \tag{15}$$

Also $JS_r L(f) = P_r f - P_0 f$. Since the entire initial germ is subtracted, this identity holds without replacing $P_0 f$ by a scalar mean.

Theorem 5.1 (Uniform reconstruction)**.** *Let $u : X \to \mathcal{O}$ be norm continuous on a topological space X, and let $K \subset X$ be compact. As A runs through the directed set of finite subsets of I ordered by inclusion,*

$$\sup_{x \in K} \|Q_A(u(x)) - L(u(x))\| \longrightarrow 0, \tag{16}$$

$$\sup_{x \in K} \sup_{r \geq 0} \|U_{A,r}(u(x)) - (P_r u(x) - P_0 u(x))\|_{\mathcal{H}} \longrightarrow 0. \tag{17}$$

These are suprema of Hilbert norms, not samplewise suprema.

Proof. For fixed A the squared error in (14) is a continuous nonnegative function of x. It decreases when A grows and converges pointwise to zero by Parseval. For a direct compactness proof, fix $\varepsilon > 0$. For each $x \in K$ choose A_x making the squared error smaller than ε^2 at x. Continuity gives a neighborhood with the same strict bound. A finite subcover of K and the union of its A_x yield one finite A_0 which works throughout K. Antitonicity gives the bound for every $A \supset A_0$. This proves (16); it is the finite-subset form of Dini's argument used in the formal proof.

For every r the contraction estimate is

$$\|S_r Q_A(f) - S_r L(f)\| \leq \|Q_A(f) - L(f)\|.$$

The same bound is independent of r. Apply the isometry J and the exact identities (15) to obtain (17). □

Proposition 5.2 (The actual Brownian-time curve)**.** *For every finite T,*

$$\sup_{0 \leq t \leq T} \|U_{A,t}(Z_t) - Z_t\|_{\mathcal{H}} \longrightarrow 0.$$

If φ is a bounded C^1 function with bounded continuous derivative, the analogous diagonal convergence holds for $\varphi(Z_t) - P_0 \varphi(Z_t)$.

Proof. Condition on the clock and use the Brownian increment isometry:

$$\|Z_t - Z_s\|_{\mathcal{H}}^2 = \mathbb{E}||W_t| - |W_s|| \le \mathbb{E}|W_t - W_s| \le |t-s|^{1/2}.$$

Thus $t \mapsto Z_t$ is norm continuous. Raw measurability gives $P_t Z_t = Z_t$. The frozen construction also proves $P_0 Z_t = 0$. One direct proof of that particular initial-germ assertion is as follows. Let $\mathcal{K}_\epsilon$ be generated by the whole clock path and by the scenery up to the clock range $R_\epsilon = \sup_{r\le\epsilon} |W_r|$. Then $\mathcal{F}_\epsilon^Z \subset \mathcal{K}_\epsilon$ and, by independence and the deterministic Brownian prediction formula conditional on the clock,

$$\mathbb{E}[Z_t \mid \mathcal{K}_\epsilon] = B_{\min(|W_t|, R_\epsilon)}, \qquad \|\mathbb{E}[Z_t \mid \mathcal{K}_\epsilon]\|^2 = \mathbb{E}\min(|W_t|, R_\epsilon) \le \mathbb{E}R_\epsilon.$$

Brownian maximal bounds and scaling give $\mathbb{E}R_\epsilon \to 0$. The initial right-germ subspace lies inside every $L^2(\mathcal{F}_\epsilon^Z)$, so its projection of Z_t has norm at most $(\mathbb{E}R_\epsilon)^{1/2}$ for every $\epsilon > 0$ and is zero. This does not claim that all initial-germ observables are constant. Theorem 5.1 with $r = t$ now proves the first assertion. Finally, $\|\varphi(Z_t) - \varphi(Z_s)\| \le \|\varphi'\|_\infty \|Z_t - Z_s\|$, and the transformed observable belongs to the raw past at t. Apply the same theorem; its initial projection remains in the formula. □

6 Recovery in the original intrinsic divergence graph

The reconstruction so far occurs in predictable-energy and observable L^2 spaces. To state the native graph limit we must specify a different, closed source operator.

Let H be the coefficient Hilbert space of either independent Gaussian source. On the joint product, let D_B and D_W be the closed Gaussian derivatives in the scenery and clock factors, with their maximal closed graph domains in $\mathcal{H}$. Let

$$\mathcal{V} = L^2(\Omega; H) \oplus L^2(\Omega; H), \qquad D_{\mathcal{O}} f = (D_B f, D_W f), \quad \operatorname{Dom} D_{\mathcal{O}} = \mathcal{O} \cap \operatorname{Dom} D_B \cap \operatorname{Dom} D_W.$$

The field norm is the square sum of the two source norms. The frozen finite-average core establishes density of this domain in $\mathcal{O}$; closedness follows from closedness of the two ambient source graphs. Define the original intrinsic divergence by the maximal adjoint $\delta_{\mathcal{O}} = D_{\mathcal{O}}^*$: a field $V \in \mathcal{V}$ belongs to its domain exactly when there is $g \in \mathcal{O}$ such that

$$\langle V, D_{\mathcal{O}} f\rangle_{\mathcal{V}} = \langle g, f\rangle_{\mathcal{H}} \quad (f \in \operatorname{Dom} D_{\mathcal{O}}); \qquad \delta_{\mathcal{O}} V = g. \tag{18}$$

Density makes g unique. This is a static two-source adjoint domain; it is not a postulated derivative in physical time.

We recall the Gaussian recovery construction in enough detail to check its domain. On one Gaussian factor, the Hermite chaos decomposition $f = \sum_{q\ge 0} f_q$ defines $Rf = D\sum_{q\ge 1} f_q/q$. Finite Hermite polynomials give

$$\|Rf\|^2 = \sum_{q\ge 1} \|f_q\|^2 / q \le \|f\|^2, \qquad \delta R f = f - \mathbb{E}f.$$

The orthogonality of differentiated chaoses proves convergence of the fields; convergence of their divergences and closedness of D^* prove membership in the original maximal adjoint. Tensor simple functions and closure give the same identities fiberwise on the joint product.

Write $C_W f = \mathbb{E}[f \mid B]$ for averaging out the clock factor and $C_B f = \mathbb{E}[f \mid W]$ for averaging out the scenery factor, and use R_B, R_W for the two fiberwise recovery maps. Independence implies $C_B C_W f = \mathbb{E}f$. Define

$$\mathfrak{R} f = (R_B(C_W f), R_W f). \tag{19}$$

The ordering is important: its two ambient divergences sum to

$$(C_W f - C_B C_W f) + (f - C_W f) = f - \mathbb{E} f.$$

Consequently, for centered $f \in \mathcal{O}$, adjointness against $\operatorname{Dom} D_{\mathcal{O}}$ proves

$$\mathfrak{R} f \in \operatorname{Dom} \delta_{\mathcal{O}}, \quad \delta_{\mathcal{O}} \mathfrak{R} f = f, \quad \|\mathfrak{R} f\|_{\mathcal{V}}^2 \le 2 \|f\|^2 . \tag{20}$$

The bound follows separately from the contraction of each fiber recovery and of C_W. No equality between the full intrinsic adjoint domain and the separate ambient adjoint domains is assumed.

Theorem 6.1 (Uniform native graph reconstruction)**.** *Set $Tz = \mathfrak{R}(Jz)$ on $\mathcal{C}_{\mathbf{e}}$. Then $Tz \in \operatorname{Dom} \delta_{\mathcal{O}}$ and $\delta_{\mathcal{O}} Tz = Jz$. For the graph metric*

$$d_{\mathrm{gr}}(V, W) = \|V - W\|_{\mathcal{V}} + \|\delta_{\mathcal{O}} V - \delta_{\mathcal{O}} W\|_{\mathcal{H}} ,$$

one has

$$d_{\mathrm{gr}}(Tz, Tw) \le 3 \|z - w\| . \tag{21}$$

Under the hypotheses of Theorem 5.1,

$$\sup_{x \in K} \sup_{r \ge 0} d_{\mathrm{gr}}(T S_r Q_A(u(x)), T S_r L(u(x))) \longrightarrow 0.$$

Proof. The output Jz belongs to $\mathcal{O}_0$ and is therefore centered, since constant functions belong to the initial germ. Apply (20). Linearity, the bound there and the isometry of J give

$$\|Tz - Tw\| \le \sqrt{2} \|z - w\| , \qquad \|\delta_{\mathcal{O}} Tz - \delta_{\mathcal{O}} Tw\| = \|z - w\| .$$

Since $1 + \sqrt{2} \le 3$, this proves (21) with the constant retained in the formal source. Combine it with the uniform input error and contractivity of every S_r. Both field values and their original divergences converge in the stated norms. □

7 Fixed stochastic coefficients as native graph tests

The fixed channels also identify every reconstruction coefficient by an original adjoint-graph test. Let

$$\mathcal{T} = \{(V, \delta_{\mathcal{O}} V) : V \in \operatorname{Dom} \delta_{\mathcal{O}}\} \subset \mathcal{V} \oplus \mathcal{O}, \quad \|(V, g)\|_{\mathcal{T}}^2 = \|V\|^2 + \|g\|^2 .$$

Closedness of the original adjoint makes this a Hilbert space. For any $f \in \mathcal{O}$ define the continuous functional

$$(\mathrm{d} f)(V, \delta_{\mathcal{O}} V) = \langle f, \delta_{\mathcal{O}} V \rangle .$$

Cauchy–Schwarz gives $\|\mathrm{d} f\|_{\mathcal{T}^*} \le \|f\|$. If $f \in \operatorname{Dom} D_{\mathcal{O}}$, original adjointness identifies it with $V \mapsto \langle D_{\mathcal{O}} f, V \rangle$ on these graph tests. The functional exists for every f; this is not an extension of the native derivative domain.

Use the already constructed bounded linear recovery $\mathfrak{R}$ of (19). For the fixed basis e_i in (13), set $V_i = \mathfrak{R} e_i$ and $\tau_i = (V_i, e_i) \in \mathcal{T}$. Equation (20) gives

$$\delta_{\mathcal{O}} V_i = e_i, \qquad \|V_i\|^2 \le 2, \qquad \|\tau_i\|_{\mathcal{T}} \le \sqrt{3}.$$

These are actual original-divergence domain elements, constructed from the fixed predictable integral outputs, rather than arbitrary formal test symbols.

Proposition 7.1. *For every $f \in \mathcal{O}$,*

$$c_i(f) = (\mathrm{d}f)(\tau_i), \qquad f - P_0 f = \sum_{i \in I} I_{e_i}((\mathrm{d}f)(\tau_i)h_i), \qquad \sum_i |(\mathrm{d}f)(\tau_i)|^2 = \|f - P_0 f\|^2 .$$

The stochastic series converges in the original observable L^2. For $f \in \operatorname{Dom} D_{\mathcal{O}}$ its coefficient can equivalently be written $\langle V_i, D_{\mathcal{O}} f\rangle$. For arbitrary f the graph-functional formula is the one asserted.

Proof. By the definition and $\delta_{\mathcal{O}} V_i = e_i$, $(\mathrm{d}f)(\tau_i) = \langle f, e_i\rangle = c_i(f)$. Substitute this equality in the fixed-channel reconstruction and Parseval identities (14). Each integral is linear in its scalar coefficient and satisfies $I_{e_i} h_i = e_i$, so these are exactly the same convergent Hilbert series, with the same squared coefficient energy. For native-domain f, apply the original adjoint relation (18) to V_i. □

There is also a useful quantitative human consequence of these same constructions. For every $f \in \mathcal{O}$,

$$\frac{\|f - P_0 f\|}{\sqrt{3}} \le \|\mathrm{d}f\|_{\mathcal{T}^*} \le \|f\| . \tag{22}$$

To prove the lower bound, take finitely supported real coefficients $\eta = (\eta_i)$ and $e_\eta = \sum_i \eta_i e_i$. Linearity and the recovery estimate on the centered observable e_η give

$$\tau_\eta := \sum_i \eta_i \tau_i = (\Re e_\eta, e_\eta), \qquad \|\tau_\eta\|_{\mathcal{T}}^2 \le 3\,\|e_\eta\|^2 = 3 \sum_i \eta_i^2 .$$

Thus $|\sum_i \eta_i c_i(f)| \le \sqrt{3}\,\|\mathrm{d}f\|\,\|\eta\|_{\ell^2}$. For a finite set A choose $\eta_i = c_i(f)$ on A and zero elsewhere. Dividing by the finite coefficient norm when it is nonzero gives $(\sum_{i\in A} |c_i(f)|^2)^{1/2} \le \sqrt{3}\,\|\mathrm{d}f\|$; the zero case is immediate. Taking increasing finite sets and Parseval proves (22). Individual bounds on $\|V_i\|$ alone would not justify this step; the bounded linear recovery on their *combinations* is what provides it.

For the actual point curve, the already proved $P_0 Z_t = 0$ and Brownian conditioning give $\|Z_t\|^2 = \mathbb{E}|W_t| = \sqrt{2t/\pi}$. Therefore

$$\frac{1}{\sqrt{3}}(2t/\pi)^{1/4} \le \|\mathrm{d}Z_t\|_{\mathcal{T}^*} \le (2t/\pi)^{1/4} \qquad (t > 0).$$

At least one fixed coefficient of Z_t is nonzero for every positive t, since otherwise Parseval would make $Z_t = 0$ in L^2. These bounds are labelled human consequences, not newly named formal theorems or sharp constants.

The graph tests are static source fields; the scalar coefficients pair the whole target with their divergences. Their stochastic integrands remain the already constructed target-indexed channels, not a new physical-time conditional-gradient formula. In particular (22) does not put the rough point target in $\operatorname{Dom} D_{\mathcal{O}}$, supply a common prescribed Brownian driver or prove observed-filtration predictable representation. It quantifies the existing generalized derivative using exact native tests.

8 Scope, antecedents and formal correspondence

The change-of-energy and associativity mechanisms belong to classical martingale integration. Métivier–Pellaumail [1] supplies the general Doléans-measure setting; Liao [2] gives a directly comparable continuous-martingale product-domain theorem. The present proof does not assume that the observed filtration is Brownian or that its terminal martingales are continuous. Its purpose

is to make the verified raw predictable domains, signed-energy quotient, original adjoint and actual Brownian-time application readable together. No new associativity principle or priority claim is made.

The uniformity proved above is $\sup_x \sup_r \|\mathrm{error}(x,r)\|_{L^2} \to 0$. It does not place a random supremum inside expectation and does not assert almost-sure uniform convergence or a càdlàg version of the moving-terminal approximants. Their coefficients depend on the terminal parameter, so the diagonal approximants are not automatically martingales. The countable representation is built from many target-indexed observable martingales. Neither a predictable representation theorem for a prescribed driver nor a temporal conditional-gradient Clark–Ocone formula follows. A native divergence representation also does not put its output in $\mathrm{Dom}\, D_{\mathcal{O}}$; rough point observations can lie outside that derivative domain.

All formal correspondences use frozen verified commit `bab688bb6140cd6316c80680095a914ea0d346db`. `IBMRightGermFiltration` identifies the genuine right-germ filtration; `IBMJointPredictableEnergyMeasure` and `IBMFullPredictableIntegral` construct the original measure and integral. `IBMPredictableRectanglePullout`, `IBMStochasticIntegratorEnergy` and `IBMStochasticProductAssociativity` supply the exact measure change and full-domain product theorem. `IBMCompletedSignedCovarianceIntegral`, `IBMMultichannelCompletion`, `IBMJointPredictableMultiplier` and `IBMCompletedBoundedAssociativity` retain all cross terms and identify the common bounded action. The countable family and uniform errors are `IBMFixedStochasticFamily` and `IBMUniformStochasticReconstruction`. `IBMNativeGaussianRecovery`, `IBMStochasticNativeDivergence` and `IBMCompletedNativeCalculus` retain the original native adjoint.

The theorem map records exact declarations and distinguishes the classical auxiliary-compensator measure proof, compactness version of Dini's argument, and elementary identical-channel example from the formal proof's organization. It also records the original finite-average density foundation for the intrinsic operator. The old five-page working note is superseded by this expanded revision, with its historical source retained in Git. No research source or active proof was modified, and no new Lean build is substituted for the frozen baseline audit.

References

[1] M. Métivier and J. Pellaumail, *A basic course on general stochastic integration*, Publications des séminaires de mathématiques et informatique de Rennes, no. 1 (1977), 1–56, especially Sections E and F. Primary text.

[2] J. Liao, *Lecture notes for stochastic analysis*, Section 5.1, Proposition 5.4, p. 145. Author's notes, accessed October 29, 2026.